

of the range of coolers that you might consider.

For a 65nm/45nm processor, you should aim for at least a 300-400 MHz stable overclock on stock cooling, while people with good cooling should aim for 600 MHz or even 1GHz.

Once you have attained a stable clock, keep it running 24x7 and notice the temperature while stress-testing; make sure it doesn't cross the maximum temperature allotted by the CPU manufacturer.

Stress testing/Prime95

Prime95 is the only efficient stress-testing program available for Linux, although the Linux version doesn't have a GUI. This test stresses the components and thus reveals any frailty. A stress-testing program stresses the CPU by upping the load to 100 per cent of the CPU cores. The processor will thus be overworked and the temperature will rise. If the overclock attained is unstable, the system might restart or stop responding while stress testing. This is a great way to check the stability of the system after overclocking.

Factors that affect overclocking

Even if you are well versed with the overclocking techniques, there is no guarantee that you will get the same results. Overclocking depends and varies with situations and the types of hardware components you use. So, I'd recommend extensive research before buying any of these components.

Let's take a quick look at what we should focus on before buying our hardware. Note that the dealers, most of the time, are uninformed and often have vested interests when they promote a product. So, do your homework on hardware peripherals before you buy them.

Conclusion

The benchmarking scenario is slowly and steadily improving in Linux. However, the clear lack of software potpourri does hurt. None of the programs tested provide an unblemished experience. For starters,

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System buying guide

Mid-High End Intel System

▪ Intel Core i5 750 (LGA 1156)	Rs 11,000
▪ Gigabyte P55 UD3R	Rs 13,000
▪ Corsair XMS3 4GB DDR3 1333 MHz RAM	~Rs 6,500
▪ Western Digital 1TB Black Edition Hard drive	Rs 5,400
▪ NVIDIA GTX 260	~Rs 10,000-11,000
▪ Samsung/LG/Lite On DVD Writer	Rs 1,200
▪ Corsair TX650 HX620 (modular) SMPS	~Rs 6,500 ~Rs 8,000
▪ Lancool PCK62 Cabinet	Rs 7,000
▪ Prolimatech MegahaleM + LGA1156 Mounting Bracket	~Rs 4,200
▪ Scythe Sflex 75cfm 120mm	Rs 500
Total: (with HX620 SMPS).....	Rs 65,300 Rs 66,800

Mid-High End Alternatives

▪ ATI Radeon HD4890.....	~Rs 10,000
▪ AMD – AMD Phenom II X4 965 Black Edition.....	Rs 13,000
▪ Intel – Intel Core i7 860	Rs 14,000-15,000

High End System

▪ Intel Core i7 920 (LGA1366)	Rs 15,000
▪ Gigabyte EAX58 UD5p	Rs 18,000
▪ Corsair XMS3 6GB DDR3 1333 Triple Channel RAM	Rs 9,000
▪ Western Digital 1Tb Black Edition Hard Drive x2	Rs 11,000
▪ ATI HD5850	Rs 17,000
▪ Samsung/LG/Lite On DVD Writer	Rs 1,200
▪ Corsair TX750W SMPS	Rs 9,000
▪ Lian Li A71F Cabinet	Rs 16,000
▪ Prolimatech MegahaleM	~Rs 3,500
▪ Scythe Sflex 75cfm 120mm	Rs 500
Total:.....	Rs 100,200

Note: All prices mentioned are courtesy deltapage.com, thelinuxware.com and primeabgb.com. The prices are quoted in INR and would vary depending on dealer discounts and product availability.

there are no overclocking software for CPUs, while there are a bunch of overclocking utilities for GPUs. The NVIDIA Control Panel does allow you to increase the memory and core speeds, while the AMD Overdrive is stagnating with no GUI on top. The lack of proper voltage and temperature measuring utilities is getting to be a sore point. Even with LM_Sensors and many other sensors, temperature monitoring isn't as precise as it should be.

Nonetheless, the benchmarking scenario is improving and will continue to do so since there is a sudden shift towards Linux. Stay tuned. 

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